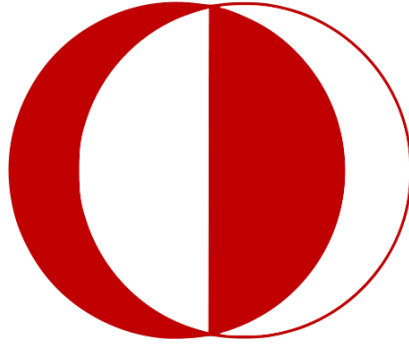




Middle East Technical University

EE568 Project-2

Motor Winding Design & Analysis

Metin Özdemir

`metin.ozdemir_01@metu.edu.tr`

github: `metinozdemir01/EE568`

1 Question 1: Baseline Integral-Slot Topology

The initial phase of the design evaluation considers a traditional double-layer integral-slot distributed winding topology with $Q = 72$ slots, $2p = 6$ poles (pole pair $p = 3$), and $m = 3$ phases. The number of slots per pole per phase is calculated as:

$$q = \frac{Q}{2 \cdot m \cdot p} = \frac{72}{2 \cdot 3 \cdot 3} = 4 \quad (1)$$

The electrical slot angle is:

$$\alpha_e = \frac{360^\circ \cdot p}{Q} = 15^\circ \quad (2)$$

1.1 Full-Pitched Winding

For a full-pitched configuration, the coil pitch is equal to the pole pitch (12 slots, or 180° electrical). The distribution factor (k_d) and pitch factor (k_p) are evaluated as follows:

$$k_d = \frac{\sin(n \cdot q \cdot \frac{\alpha_e}{2})}{q \cdot \sin(n \cdot \frac{\alpha_e}{2})} = \frac{\sin(1 \cdot 4 \cdot 7.5^\circ)}{4 \cdot \sin(1 \cdot 7.5^\circ)} = 0.957 \quad (3)$$

$$k_p = \sin\left(\frac{180^\circ}{2}\right) = 1 \quad (4)$$

$$k_w = k_d \cdot k_p = 0.957 \cdot 1 = 0.957 \quad (5)$$

Where n is the order of harmonics. For the spatial harmonics, the third and fifth winding factors drop significantly. The third harmonic distribution factor evaluates to $k_{d3} = 0.653$ and the pitch factor is $k_{p3} = -1$, leading to a winding factor of $k_{w3} = -0.653$. The fifth harmonic distribution factor evaluates to $k_{d5} = 0.205$ and the pitch factor is $k_{p5} = 1$, yielding a winding factor of $k_{w5} = 0.205$.

1.2 Short-Pitched Winding (11/12)

To suppress the spatial harmonics and reduce end-winding length, an 11/12 short-pitched winding is analyzed. The new coil pitch is 165° electrical.

$$k_p = \sin\left(\frac{165^\circ}{2}\right) = 0.991 \quad (6)$$

$$k_w = k_d \cdot k_p = 0.957 \cdot 0.991 = 0.948 \quad (7)$$

The harmonic winding factors for the short-pitched design are similarly attenuated. The third harmonic yields $k_{d3} = 0.653$ and $k_{p3} = 0.991$, resulting in $k_{w3} = 0.647$. The fifth harmonic yields $k_{d5} = 0.205$ and $k_{p5} = 0.991$, resulting in $k_{w5} = 0.203$.

1.3 Winding Diagrams

The following tables illustrate the phase band layouts for the first pole pair (Slots 1–24) of the 72-slot machine. Because $q = 4$, each phase group spans 4 consecutive slots. Table 1 presents the single-layer full-pitch configuration, while Table 2 details the double-layer 11/12 short-pitched configuration where the bottom layer shifts by one slot.

Table 1: Table 1: Single-Layer Full-Pitch Winding Diagram (Slots 1–24)

Slot Numbers	1–4	5–8	9–12	13–16	17–20	21–24
Phase	A+	C-	B+	A-	C+	B-

Table 2: Table 2: Double-Layer 11/12 Short-Pitched Winding Diagram (Slots 1–24)

Slot Numbers	1–4	5–8	9–12	13–16	17–20	21–24
Top Layer	A+	C-	B+	A-	C+	B-
Bottom Layer	A+ (1–3) C- (4)	C- (5–7) B+ (8)	B+ (9–11) A- (12)	A- (13–15) C+ (16)	C+ (17–19) B- (20)	B- (21–23) A+ (24)

2 Question 2: Fractional-Slot Winding Design

To achieve high efficiency and compact end-windings, a 15-slot, 14-pole (15/14) fractional-slot concentrated winding (FSCW) topology was chosen. For $Q = 15$ and $p = 7$:

$$q = \frac{Q}{2 \cdot m \cdot p} = \frac{15}{2 \cdot 3 \cdot 7} = \frac{5}{14} \quad (\text{or } 5/7 \text{ slots per pole pair per phase}) \quad (8)$$

The electrical slot angle is calculated as:

$$\alpha_e = 360^\circ \cdot \frac{p}{Q} = 360^\circ \cdot \frac{7}{15} = 168^\circ \quad (9)$$

Based on the star of slots, the fundamental winding factor for this fractional configuration is evaluated as:

$$k_w = \frac{A \cdot (1 + 2 \sin(66^\circ) + 2 \sin(78^\circ))}{5A} = 0.956 \quad (10)$$

Harmonic winding factors evaluate to $k_{w3} = 0.647$ for the third harmonic and $k_{w5} = 0.02$ for the fifth harmonic.

2.1 Phasor Diagrams and Winding Factor

The fundamental winding factor is visually verified through the star of slots and the phase vector diagrams. The star of slots for the 15-slot machine is constructed using an electrical slot angle of 168° , generating 15 distinct voltage phasors as illustrated in Figure 1. This diagram guides the optimal assignment of positive and negative coil sides to maximize the fundamental magnetomotive force.

For Phase A, the specific slot assignments create a symmetrical distribution of five vectors centered around the horizontal axis. Figure 2 demonstrates the tip-to-tail vector addition of these specific phase components, which are oriented at 0° , 24° , -24° , 12° , and -12° . The geometric resultant vector stretches along the real axis. Dividing the magnitude of this resultant vector by the total number of coils per phase yields the fundamental winding factor of 0.956. This graphical method physically proves the analytical calculation established earlier in this section.

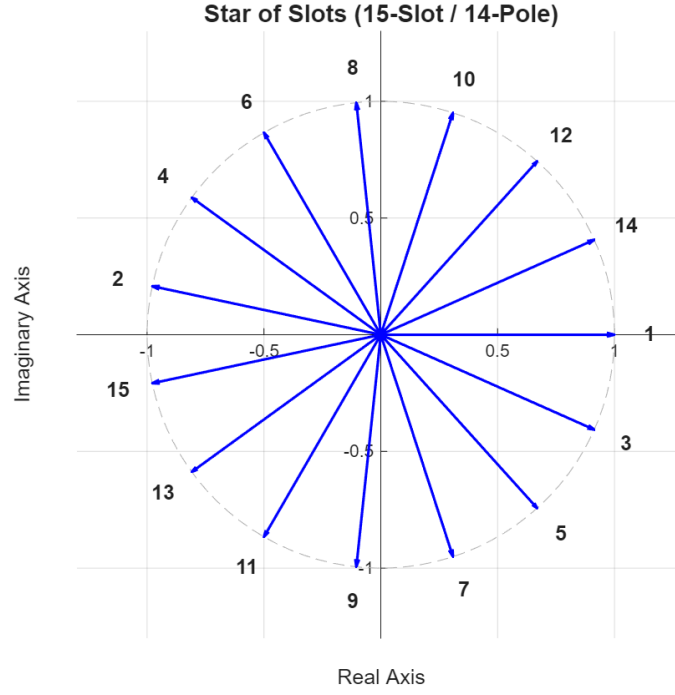


Figure 1: Star of slots for $Q=15$, $2p=14$.

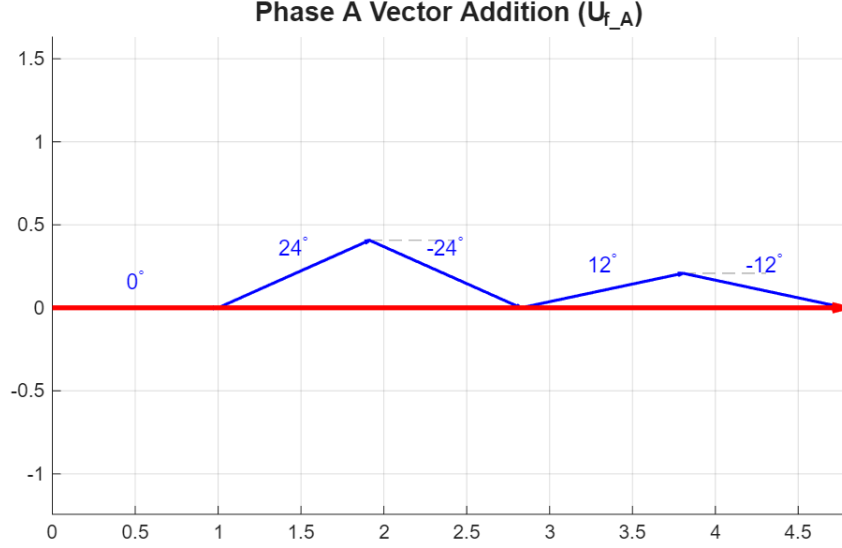


Figure 2: Phase A tip-to-tail vector addition (U_{f_A}).

2.2 Winding Layout

Table 3 illustrates the complete layer mapping across the 15 stator slots. The phase distribution features distinct phase boundary slots where conductors from two different phases share the same space.

Table 3: 15-Slot / 14-Pole Double-Layer Winding Layout

Slot	Top Layer	Bottom Layer	Slot	Top Layer	Bottom Layer
1	A+	A+	9	B+	B+
2	A-	A-	10	B-	B-
3	A+	C-	11	B+	B+
4	C+	C+	12	B-	B-
5	C-	C-	13	B+	A-
6	C+	C+	14	A+	A+
7	C-	C-	15	A-	A-
8	C+	B-			

2.3 Verification of Winding Validity and Balance

The double-layer winding layout presented in Table 3 satisfies the fundamental criteria for both validity and symmetrical three-phase balance.

To ensure the winding is valid and does not produce circulating currents, the net magnetomotive force for each phase must naturally sum to zero. Examining Phase A, the positive coil sides (A+) are positioned twice in Slot 1, once in Slot 3, and twice in Slot 14, yielding exactly 5 positive conductors. The negative coil sides (A-) are positioned twice in Slot 2, once in Slot 13, and twice in Slot 15, yielding exactly 5 negative conductors. Because the total number of inward and outward paths are strictly equal, the net phase MMF is completely balanced.

Furthermore, the three phases are symmetrically identical but physically displaced across the stator perimeter by exactly 5 slots. Given the electrical slot angle of $\alpha_e = 168^\circ$, a 5-slot physical shift corresponds to an electrical phase shift of:

$$\theta_{shift} = 5 \text{ slots} \times 168^\circ/\text{slot} = 840^\circ \text{ electrical} \quad (11)$$

Because an angle of 840° is geometrically equivalent to 120° ($840 \pmod{360} = 120^\circ$), the phase bands perfectly satisfy the mandatory 120° electrical displacement required for a balanced three-phase system.

3 Question 3: Analytical Sizing and Magnetic Circuit

The machine was analytically sized for a rated torque of 11.3 Nm at 1500 RPM, constrained by a 48 V DC bus limit.

3.1 Geometry and Effective Air Gap

Given an outer stator radius $R_{so} = 50$ mm and an outer rotor radius $R_{ro} = 31$ mm, the stator inner radius is established considering the magnet thickness ($l_m = 4$ mm) and physical air gap ($g = 1$ mm):

$$R_{si} = R_{ro} + l_m + g = 31 + 4 + 1 = 36 \text{ mm} \quad (12)$$

To account for the slot openings, Carter's coefficient (k_c) is calculated. With a tooth pitch $\tau_s = \frac{2\pi(36)}{15} = 15$ mm and a slot opening $b_1 = 0.45\tau_s \cong 6.75$ mm:

$$\delta = g + \frac{l_m}{\mu_r} = 1 + \frac{4}{1.05} = 4.81 \text{ mm} \quad (13)$$

$$\kappa = \frac{b_1/\delta}{5 + b_1/\delta} = \frac{6.75/4.81}{5 + 6.75/4.81} = 0.22 \quad (14)$$

$$k_c = \frac{15}{15 - (0.22 \cdot 6.75)} \cong 1.11 \quad (15)$$

Thus, the effective air gap clearance is $\delta_e = k_c \cdot g = 1.11$ mm.

3.2 Magnetic Loading

Assuming an effective axial length $L_{eff} = L_s + 2g = 100 + 2 = 102$ mm, the peak air-gap flux density is:

$$B_{peak} \approx \frac{B_r}{1 + \mu_r \cdot \frac{g \cdot k_c}{l_m}} = \frac{1.3}{1 + 1.05 \cdot \frac{1 \cdot 1.11}{4}} \cong 1.0 \text{ T} \quad (16)$$

Applying a magnet-pole arc ratio of 0.89 ($\sim 160^\circ/180^\circ$), the average magnetic loading is:

$$\bar{B} = B_{peak} \cdot 0.89 = 1.0 \cdot 0.89 = 0.89 \text{ T} \quad (17)$$

The fundamental flux per pole is therefore:

$$\Phi = B_{avg} \cdot \frac{2\pi R_{si} L_{eff}}{2p} = 0.89 \cdot \frac{2\pi(0.036)(0.102)}{14} = 1.466 \text{ mWb} \quad (18)$$

3.3 Electric Loading and Sizing

To maintain stator core flux densities near the 1.4 T limit, roughly 40% of the total slot area is assigned to copper, yielding $A_{cu} \approx 60.6 \text{ mm}^2$ per slot. Selecting a current density of $J = 5 \text{ A/mm}^2$:

$$I_{total} = J \cdot A_{cu} = 5 \cdot 60.6 = 303 \text{ A}_{rms} \quad (19)$$

The specific electric loading ($\bar{A}$) evaluates to:

$$\bar{A} = \frac{15 \cdot 303}{\pi \cdot 2 \cdot 0.036} = 20,093 \text{ A}_{rms}/\text{m} \quad (20)$$

Using these loadings, the shear stress (σ) and electromagnetic torque (T) are:

$$\sigma = \frac{1}{\sqrt{2}} \cdot k_w \cdot \bar{A} \cdot B_{peak} = \frac{1}{\sqrt{2}}(0.956)(20093)(1.0) = 13,582 \text{ N/m}^2 \quad (21)$$

$$T = 2\pi R_{si}^2 L_{eff} \sigma = 2\pi(0.036)^2(0.102)(13582) = 11.28 \text{ Nm} \quad (22)$$

3.4 Phase Turns Calculation

For a 48 V DC bus, the available phase voltage (assuming star connection) is $V_{rms} = \frac{48}{\sqrt{3}} = 27.7 \text{ V}$. At 1500 RPM, the electrical frequency is $\omega_e = 7 \cdot 157 = 1099 \text{ rad/s}$. The required number of series turns per phase is:

$$N_{ph} = \frac{V_{rms}}{\omega_e \cdot k_w \cdot \Phi} = \frac{27.7}{1099 \cdot 0.956 \cdot (1.466 \times 10^{-3})} \cong 18 \text{ turns} \quad (23)$$

4 Question 4: Finite Element Analysis Validation

In this section, the analytical design of the 15-slot, 14-pole PMSM is validated using 2D finite element analysis in COMSOL Multiphysics. All simulations are performed under open-circuit conditions to precisely evaluate the core geometry, magnetic saturation behavior, and parasitic reluctance forces.

4.1 Magnetic Flux Density Distribution

The magnetic flux density distribution across the machine core verifies the structural sizing of the magnetic circuit. As observed in the overall surface plot, the stator back-core and teeth operate efficiently at approximately 0.9 T. This aligns perfectly with the preliminary analytical assumption of $\bar{B} = 0.89$ T and remains comfortably below the strict 1.4 T saturation limit.

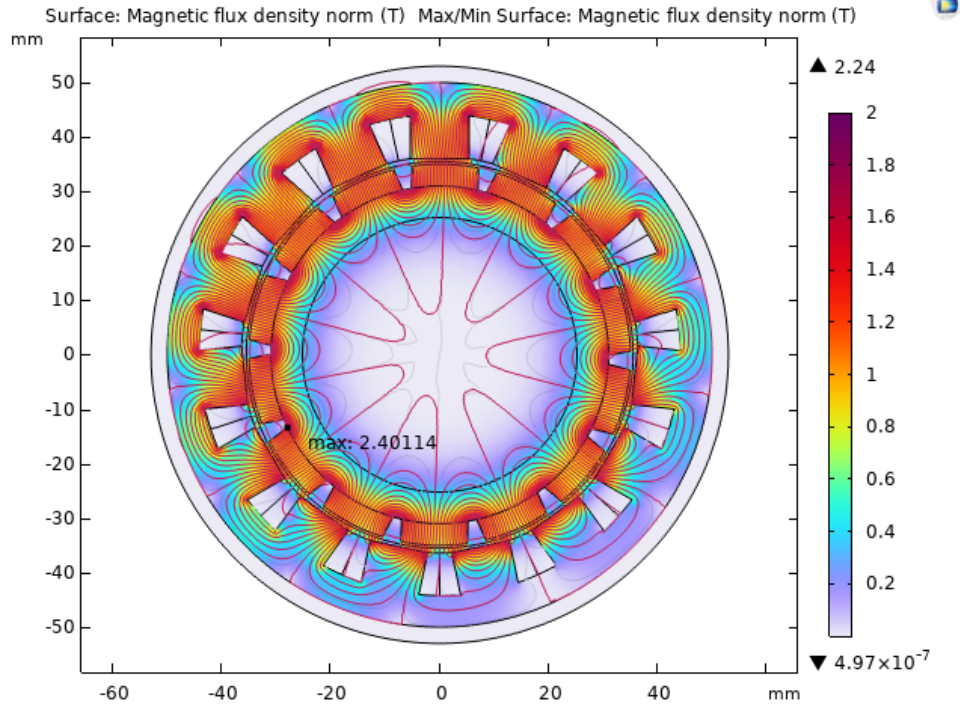


Figure 3: Surface plot of the magnetic flux density norm across the stator and rotor cores.

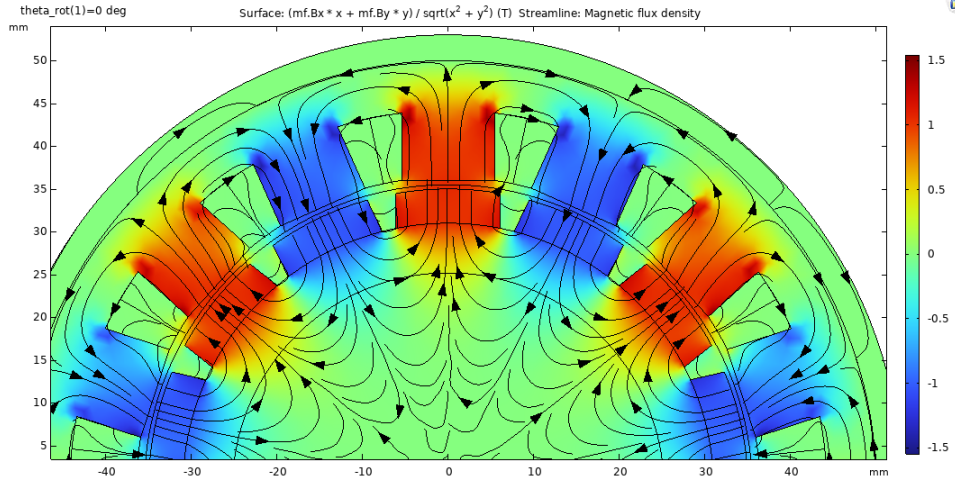


Figure 4: Streamline plot illustrating the directional flow of the magnetic flux density.

4.2 Leakage Flux Paths

A detailed examination of the magnetic vector potential contours reveals several critical leakage paths within the machine. The most prominent is the magnet-to-magnet leakage located at the narrow rotor bridges separating adjacent permanent magnets. These regions exhibit severe, localized magnetic saturation, reaching peak values of approximately 2.4 T. By driving the relative permeability of the iron bridge down to levels comparable to air, the magnet-to-magnet leakage flux is successfully choked, forcing the majority of the useful magnetic field to cross the air gap.

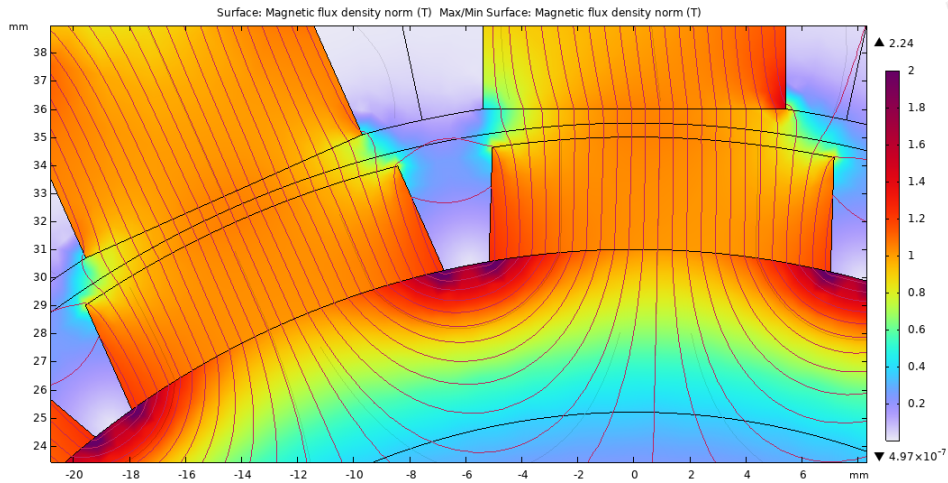


Figure 5: Detailed view of magnet-to-magnet leakage and saturation at the rotor bridges.

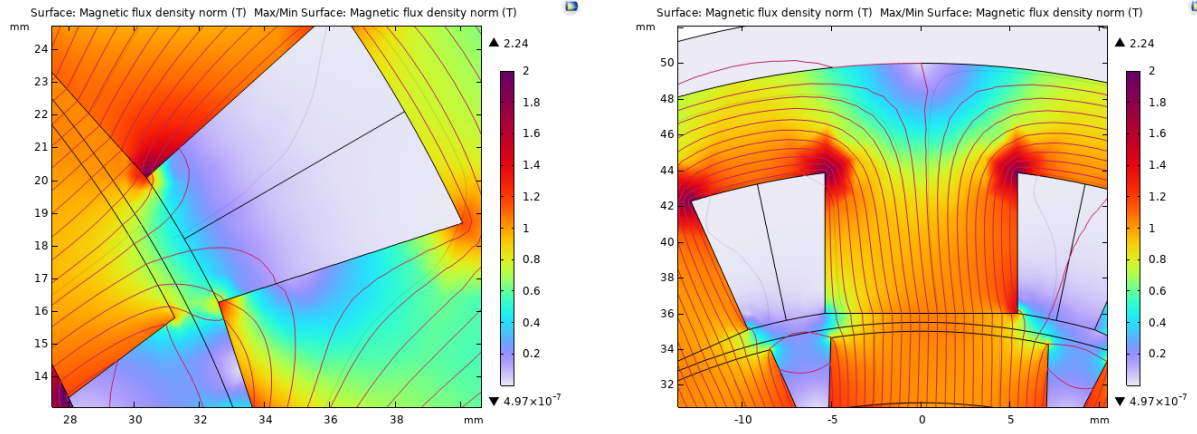


Figure 6: Localized views of tooth-to-magnet leakage (left) and back-core leakage patterns (right).

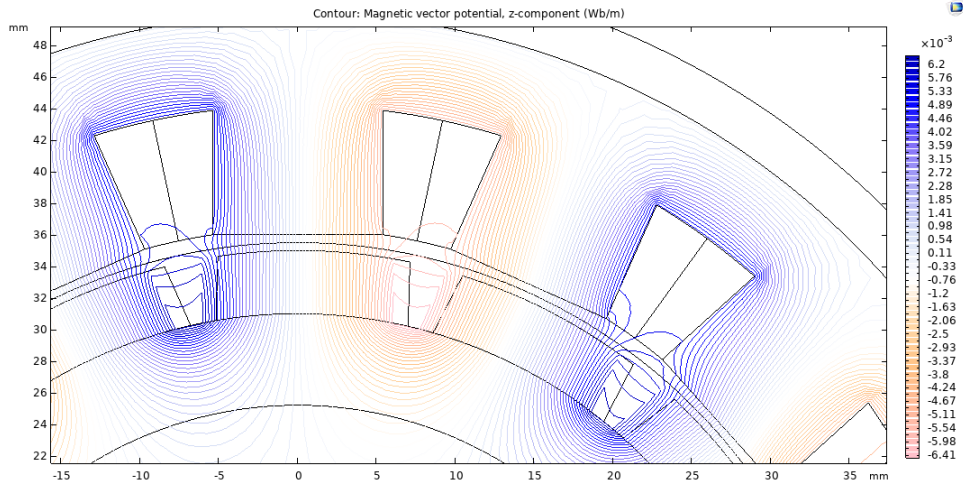


Figure 7: Overall contour map of the magnetic vector potential highlighting the full distribution of leakage paths.

4.3 Air-Gap Flux Density

The radial air-gap flux density was evaluated along a continuity boundary located at the center of the air gap at a 35.5 mm radius. The finite element solver yields a peak air-gap flux density of approximately 1.05 T, verifying the 1.007 T peak predicted during the 1D analytical sizing phase. The waveform exhibits localized jagged dips at the peaks of each magnetic pole, representing the physical fringing effects caused by the stator slot openings modeled by Carter's coefficient.

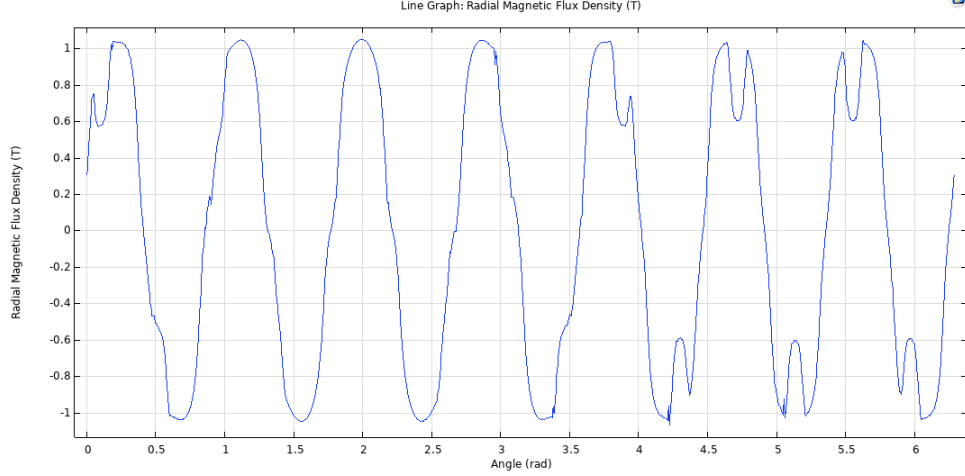


Figure 8: Radial air-gap flux density distribution over a full mechanical rotation.

4.4 Flux-Per-Pole Comparison

To determine the fundamental flux per pole (Φ_{FEA}), the radial flux density was mathematically integrated over exactly one pole pitch and scaled by the effective axial length:

$$\Phi_{FEA} = L_{eff} \int_0^{\pi/7} B_r \cdot r_{gap} d\theta = 1.085 \text{ mWb} \quad (24)$$

This simulated value is lower than the analytically calculated value of 1.466 mWb. This discrepancy is expected, as the analytical model assumes an ideal rectangular flux distribution and ignores the substantial magnet-to-magnet and tooth-tip leakage verified in the contour analysis.

4.5 Cogging Torque Analysis

The cogging torque of the machine was evaluated by computing the Maxwell surface torque on the rotor. The finite element waveform shows a peak-to-peak cogging torque ripple of approximately $T_{cog} = 0.24 \text{ Nm}$. When compared to the rated electromagnetic torque of $T_{rated} = 11.28 \text{ Nm}$, this cogging torque represents a ripple of only 2.1%. Maintaining a high least common multiple (210) between the stator slots and rotor poles neutralizes the magnetic reluctance forces, ensuring smooth rotational operation.

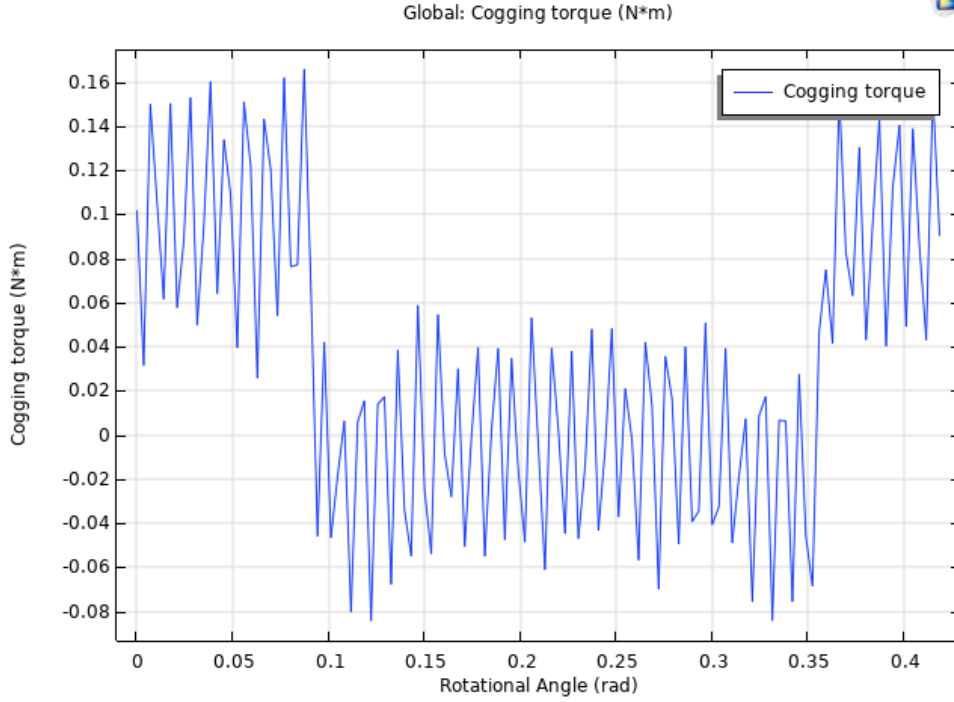


Figure 9: Cogging torque waveform evaluated over the mechanical angle.

5 Conclusion

This report presented the comprehensive design, analytical sizing, and finite element validation of a three-phase Permanent Magnet Synchronous Machine. The design process began with the evaluation of a baseline 72-slot, 6-pole integral-slot topology. While suitable for generating sinusoidal MMF, the distributed nature of the windings introduced significant disadvantages regarding end-turn copper volume and phase resistance, which are detrimental to a compact, highly efficient motor design operating within a strict 48 V DC bus constraint.

To overcome these physical limitations, the architecture was transitioned to an optimized 15-slot, 14-pole fractional-slot concentrated winding configuration. This specific slot-pole combination was selected for its high least common multiple, which acts as an inherent spatial filter to drastically reduce reluctance forces without the need for complex mechanical skewing. The analytical magnetic circuit was meticulously sized to deliver the required 11.3 Nm of rated torque at 1500 RPM while maintaining the stator iron below a 1.4 T saturation boundary.

Subsequent 2D finite element analysis in COMSOL Multiphysics rigorously validated the analytical framework. The simulations confirmed the stator core operates efficiently near the targeted 0.9 T average loading. Furthermore, the analysis successfully demonstrated the deliberate localized saturation of the rotor bridges at 2.4 T, effectively choking magnet-to-magnet leakage and maximizing active air-gap flux. Finally, the simulated cogging torque ripple was confirmed to be an exceptionally low 2.1 percent of the rated torque, proving that the selected 15/14 fractional-slot design achieves smooth, high-density performance perfectly aligned with the project requirements.